

Project Title: **Draya**

Team Name: **Draya Team**

Target Industry: **EdTech / AI in Education**

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1. The Core Problem (The "Why")

Teachers spend a lot of time creating exams, tracking students manually, and analyzing performance results. Students also study passively without understanding their weak areas, while parents cannot clearly monitor their child's academic progress.

This problem is especially common in tutoring centers and private educational academies that manage large numbers of students and exams manually.

Most existing learning platforms only provide content uploading and online exams without AI-driven performance analysis, summaries, or personalized insights.

2. The Solution & Value Proposition (The "What")

Draya is an AI-powered web and mobile learning platform that helps teachers upload lesson materials, select exam type and difficulty, review AI-generated questions before publishing, and receive automated analytics after students complete exams. Students can access lessons, take exams, and receive AI-generated summaries and performance insights.

The platform also provides topic-level weakness analysis, parent reports, and a Q&A communication channel between teachers and students.

Key Benefit:

Saving teachers' time while improving student learning through AI-generated insights and automated academic analysis.

3. The AI Trinity Strategy (The "How")

• LLM (Intelligence)

Claude Sonnet 4.6. Used exclusively for Agent 1 (Exam Builder), Agent 2 (Grader), and Agent 3 (Report Generator). By centralizing Sonnet 4.6, we ensure maximum accuracy for curriculum extraction and structured JSON evaluation, while simultaneously leveraging its advanced semantic reasoning to generate nuanced, empathetic academic and weakness reports. Claude also provides industry-leading Arabic language capabilities, ensuring high-quality, grammatically accurate generation tailored for the Egyptian educational ecosystem.

Its primary roles are:

- Generating exams and quizzes
- Summarizing lessons
- Analyzing feedback
- Generating performance insights and recommendations

• RAG (Knowledge)

- Uploaded educational content will first be processed using document parsing before retrieval and analysis.
- Course materials
- Student exam results
- Feedback and analytics data
- **Vector Database:** Qdrant Cloud (Free Tier), accessed via its official C# SDK. This provides instant setup, security, and an isolated semantic search engine that keeps heavy AI vector math away from our operational MSSQL database.
- **Embedding Model:** Nova Multimodal Embeddings, used to convert Arabic and English lesson materials into dense vectors.
- **Reranking Engine:** Cohere Rerank 3.5.
- **Chunking Strategy:** Fixed-size chunking with overlap. Uploaded PDFs are parsed and split into chunks of 1,000 characters with a 150-character overlap. Each chunk is tagged with C# metadata classes containing SubjectId, LessonId, and DifficultyLevel to enable metadata pre-filtering inside Qdrant before running vector similarity matching.

This allows the AI to generate context-aware exams and personalized recommendations instead of generic responses.

• Agents (Action)

The system will use specialized AI agents with clearly defined workflows:

- **Agent 1: Exam Builder**
Retrieves lesson content through the RAG pipeline, generates curriculum-aware questions based on the selected topic and difficulty level, and formats the final exam for teacher review. Specifically, this agent decides the exact text snippets to extract from the RAG chunks and determines the correct alternative options (distractors) for multiple-choice questions based on the targeted lesson difficulty.

- **Agent 2: Grader**
Evaluates student responses using predefined marking criteria and generates performance metrics based on accuracy and topic understanding.
 - **Agent 3: Report Generator**
Aggregates student performance data, identifies weak learning areas and trends, and generates structured summaries and reports for teachers and parents. Specifically, this agent decides which specific educational topics fall below proficiency benchmarks to categorize them as a "weak learning area" in the outputted JSON schema. Topic-level weakness analysis and parent reports are drafted by this agent but must be approved by a human (the teacher) before becoming visible to parents.
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4. Market & Competition

Primary User

- Teachers in tutoring centers and private academies.

Secondary Users

- Students
- Parents

Geographic Focus: Egypt

Language Support: Arabic-first platform with English support.

• The Advantage

Unlike traditional LMS platforms like Moodle that only provide content management and online exams, Draya integrates AI directly into the learning and evaluation process.

The platform offers:

- Adaptive AI-generated exams based on lesson content and difficulty level.
- Curriculum-aware RAG that generates questions and summaries from uploaded lesson materials and course documents.
- Automated student performance analysis to detect weak topics and learning patterns.
- AI-generated reports and insights for teachers and parents
- Arabic-first educational support designed for Egyptian learning environments.
- Analytics dashboards and AI-powered feedback analysis unavailable in most traditional LMS platforms.

This reduces manual work for teachers, improves learning quality for students, and provides more personalized academic insights.

5. Execution Plan (High Level)

• Core Stack

Frontend: Angular and Flutter

Backend: ASP.NET Core Web API

Database: MSSQL Server

AI Orchestration: Microsoft Semantic Kernel (integrates natively with the ASP.NET Core backend via C# dependency injection) orchestrating Claude Sonnet 4.6 via the Anthropic API.

• Risk Factor

The biggest technical challenge is not only maintaining high-quality RAG data but also ensuring the reliability and fairness of AI-generated outputs.

Key risks include:

- Hallucination prevention in generated exams, summaries, and reports
- Ensuring grading accuracy and evaluation fairness
- Reducing AI bias in analysis and recommendations
- Preventing cheating and misuse during online exams
- Maintaining privacy and security of student educational data
- Managing LLM API costs while maintaining scalable response times and reliable AI performance.

Another challenge is managing different educational content formats such as PDFs, and uploaded files while maintaining accurate AI understanding and analysis.

6. Technical Architecture, Governance & Compliance

• Agent Framework

We will use Microsoft Semantic Kernel as the agent framework. Since Draya's backend is built on ASP.NET Core Web API, Semantic Kernel is the native industry choice: it integrates seamlessly with C# dependency injection, allows Agents to be defined as strongly-typed C# services, and natively supports plugins, chat completion, and prompt pipelines without the overhead of Python-based tools.

• AI Observability & Monitoring

Tool: Langfuse. The Langfuse SDK is integrated into the ASP.NET Core API layer. Every time a teacher generates an exam or a student submits an assessment for grading, the call is wrapped in a Langfuse trace, allowing the team to track exact prompt payloads, token usage costs, latency metrics, and catch hallucination errors directly from a visual dashboard.

• Deployment, Containerization & CI/CD

- **Hosting:** Azure App Service (Angular frontend and ASP.NET Core API) combined with a managed Azure SQL Database for MSSQL.

- **Docker:** The ASP.NET Core Web API and Angular frontend are containerized using multi-stage Dockerfile configurations to ensure identical development and production environments.
- **Automated Testing:** Unit tests in xUnit cover repository patterns, DTO mappings, and Semantic Kernel plugin definitions.
- **CI/CD Pipeline:** A GitHub Actions workflow triggers on every push to the main branch. It runs the xUnit test suite, builds the Docker images, pushes them to Azure Container Registry (ACR), and automatically deploys them to Azure App Service.

• Student Data Privacy & Compliance

Student data privacy is maintained by ensuring real student identities never reach third-party AI endpoints:

- **PII Anonymization Pipeline:** a middleware service in the ASP.NET Core backend intercepts requests destined for Claude, replaces student names/emails with internal database GUIDs, and re-maps the returned JSON payloads back to the real student profiles inside the secure MSSQL database.
- **Encryption:** all communications are forced over HTTPS.
- **Authentication:** handled securely via JWT (JSON Web Tokens) managed by ASP.NET Core Identity.
- **RBAC:** Role-Based Access Control enforces strict permission levels — parents can only read data linked to their specific child's ID, students cannot access draft exams, and teachers own exclusive write permissions for their tutoring center's classrooms.

• Human-in-the-Loop & Hallucination Prevention

- **Human-in-the-Loop:** Yes, strictly. Topic-level weakness analysis and parent reports are drafted by the AI but must be approved by a human (the teacher). In the MSSQL schema, the ParentReport table features an IsApproved boolean column (defaulting to false); the report only becomes visible on the Flutter/Web parent view once the teacher reviews and clicks “Approve” on their dashboard.
- **Hallucination Guardrails:** model temperature is locked to 0.0 for grading and exam generation, and Semantic Kernel system prompts enforce a strict guardrail: “Answer ONLY using the provided textbook context. If the context does not contain the information, reply with ‘DATA_UNAVAILABLE.’ Do not invent details.”

• Cheating Prevention & Grading Fairness

Multiple mechanisms ensure academic integrity:

- Randomized question order
- Question pools
- Time limits
- Browser tab switching detection

- Copy-paste restrictions
- Teacher approval before publishing generated exams

To improve grading fairness:

- Objective questions are graded deterministically
- Teachers can review and override AI scores
- Confidence scores are displayed for every AI evaluation

• **Cost vs. Benefit Analysis**

Traditional Approach (Teaching Assistants)	Draya AI
Requires multiple TAs as student numbers grow	Scales to thousands of students with minimal additional cost
Hours to create and grade exams	Generates and grades exams in minutes
Manual performance analysis	Automatic topic-level weakness analysis
Manual parent reports	AI-generated reports with teacher review
Ongoing salary and training costs	Pay-per-use AI APIs with optimized token usage, caching, and RAG to reduce costs